CO-NORMALIZER ORTHOGONAL JOIN REDUCTION AND RATIONAL CHERN–SIMONS VOLUME ON ONE-ANGLE SPHERICAL BAR CYCLES

ARGUS OPEN RESEARCH PROJECT

Abstract. A conjecture in low-dimensional topology asks whether the $\mathrm{SU}(2)$ Chern–Simons invariant of every flat connection on a closed 3-manifold is rational modulo 1. For the one-angle spherical bar-cycle construction in the Bloch/scissors congruence group, we prove that the co-normalizer flatness constraint forces an orthogonal spherical join, and that vanishing of the Dehn invariant on the resulting cycle forces its normalized Chern–Simons volume to be rational.

1. Main Rationality Reduction

Theorem 1.1. *In the one-angle spherical bar-cycle family, imposing the co-normalizer holonomy constraint decomposes every admissible tetrahedron cycle as an orthogonal spherical join; tensoring the Schlafli/Dehn invariant over $\mathbb{Q}$ shows that whenever the Dehn invariant vanishes, the dihedral angles are rational multiples of π and the normalized Chern–Simons invariant lies in $\mathbb{Q}/\mathbb{Z}$.*

Date: September 5, 2026.
2020 *Mathematics Subject Classification.* Primary 57K32; Secondary 58J28, 19E20.